

SEOUL SINYONGSAN, Korea, Republic of

WMO: 471170

Lat: 37.533N

Lon: 126.967E

Elev: 95

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 82-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	13.9	-3.9	4.5	15.6	-0.4	5.4	18.2	15.9	36.3	13.8	33.2	3.2	330	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	91.7	77.3	89.6	76.0	87.4	75.3	79.4	87.9	78.4	86.1	77.4	84.2	5.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.1	83.0	76.5	138.6	82.4	75.3	133.0	81.5	43.0	87.7	41.9	86.3	40.9	83.8	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	11.7	10.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	27.6	32.9	43.2	56.7	64.9	73.5	79.0	79.8	71.2	60.2	46.9	34.7
	DBStd	18.81	7.29	8.24	6.29	6.99	4.91	4.51	4.59	4.28	5.37	6.39	7.45	7.88
	HDD50	2040	695	478	223	19	1	0	0	0	0	9	142	473
	HDD65	4716	1160	898	677	260	61	3	0	0	9	167	543	938
	CDD50	4232	0	0	11	221	462	705	898	923	636	327	49	0
	CDD65	1433	0	0	0	12	58	258	433	458	195	19	0	0
	CDH74	11147	0	0	0	81	335	1642	3754	4208	1067	60	0	0
	CDH80	3600	0	0	0	9	44	458	1345	1501	242	1	0	0
(14) Wind	WSAvg	3.8	3.4	3.9	4.8	4.7	4.2	3.8	4.0	3.9	3.4	2.9	3.5	3.0
(15) Precipitation	PrecAvg	54.8	0.7	1.0	1.7	3.3	3.9	5.3	15.6	12.4	6.0	2.0	2.0	0.9
	PrecMax	92.8	2.5	4.3	4.9	13.3	11.5	19.6	44.5	48.7	26.5	8.4	6.5	2.8
	PrecMin	30.0	0.0	0.0	0.1	0.3	0.3	0.6	4.5	2.6	0.2	0.0	0.3	0.1
	PrecStd	13.9	0.6	0.9	1.1	2.6	2.5	4.1	8.2	8.7	5.4	1.6	1.3	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	55.1	64.4	80.7	84.1	91.3	95.4	95.1	89.7	78.5	66.2	55.5
		MCWB	39.7	45.6	49.4	61.3	64.0	71.2	79.6	78.6	75.1	63.1	56.3	47.4
	2%	DB	42.8	50.3	60.5	75.5	80.6	87.9	92.9	91.8	85.9	75.4	62.8	50.4
		MCWB	37.6	43.0	47.6	60.7	63.1	70.5	78.8	77.7	73.0	62.9	54.5	44.5
	5%	DB	40.7	47.2	56.9	71.8	77.3	84.7	89.6	89.7	82.5	72.7	60.4	48.4
		MCWB	36.1	40.1	46.1	57.5	61.3	69.5	77.4	76.7	70.7	61.0	52.3	42.8
	10%	DB	37.7	44.2	53.5	68.3	75.1	82.4	86.4	87.5	79.2	69.8	57.4	46.2
		MCWB	33.6	38.0	44.6	55.7	61.0	68.5	75.9	76.4	69.0	59.8	50.7	40.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	48.1	52.1	67.2	69.5	75.4	81.3	80.7	77.6	66.7	59.3	49.8
		MCDB	44.6	52.0	60.8	74.2	76.5	84.1	91.1	88.9	84.3	73.0	63.9	53.7
	2%	WB	38.4	43.8	49.3	62.4	67.0	73.4	79.9	79.4	75.7	64.8	56.7	46.4
		MCDB	41.8	49.3	57.2	73.4	74.7	82.6	89.2	87.5	81.8	71.4	61.1	49.6
	5%	WB	36.4	40.4	47.0	59.5	64.8	71.8	78.7	78.6	74.0	63.2	53.8	43.2
		MCDB	39.8	46.5	55.2	70.0	72.7	80.8	86.8	86.4	79.7	69.6	58.5	47.8
	10%	WB	34.1	38.0	44.9	57.1	62.9	70.3	77.6	77.6	72.0	61.4	51.3	40.9
		MCDB	37.3	43.3	52.2	66.2	71.5	78.5	84.6	84.7	77.6	68.3	56.6	45.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.3	15.7	18.7	17.1	15.4	12.0	12.5	14.7	16.5	14.7	13.3
		MCDBR	13.9	16.2	20.1	22.3	21.1	19.2	15.7	15.3	16.6	18.9	16.7	14.5
		MCWBR	10.8	11.0	11.7	11.2	9.4	8.1	6.2	6.0	7.5	9.1	10.4	10.7
	5% WB	MCDBR	13.2	14.7	18.8	19.7	17.1	15.9	13.2	12.9	13.9	16.2	14.5	13.1
		MCWBR	11.3	10.6	12.1	11.1	9.4	7.8	5.8	5.9	7.6	9.4	11.0	11.0
(40) Clear-Sky Solar Irradiance	taub		0.441	0.496	0.579	0.626	0.638	0.710	0.678	0.589	0.516	0.509	0.489	0.439
	taud		1.934	1.839	1.700	1.650	1.674	1.597	1.711	1.913	2.036	1.934	1.925	1.963
	Ebn at Noon		219	223	218	218	219	203	208	225	234	217	202	209
	Edh at Noon		45	56	70	78	77	84	74	59	50	50	45	41
(44) All-Sky Solar Radiation	RadAvg		793	1037	1354	1577	1774	1658	1291	1378	1311	1137	774	696
	RadStd		49	82	135	110	164	171	197	195	143	80	75	48

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+90

Nomenclature: See separate page